
Six Week Review

Fall Prevention

Exoskeleton Clinic

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Our Team



Chase Quijano



Preston Haddon



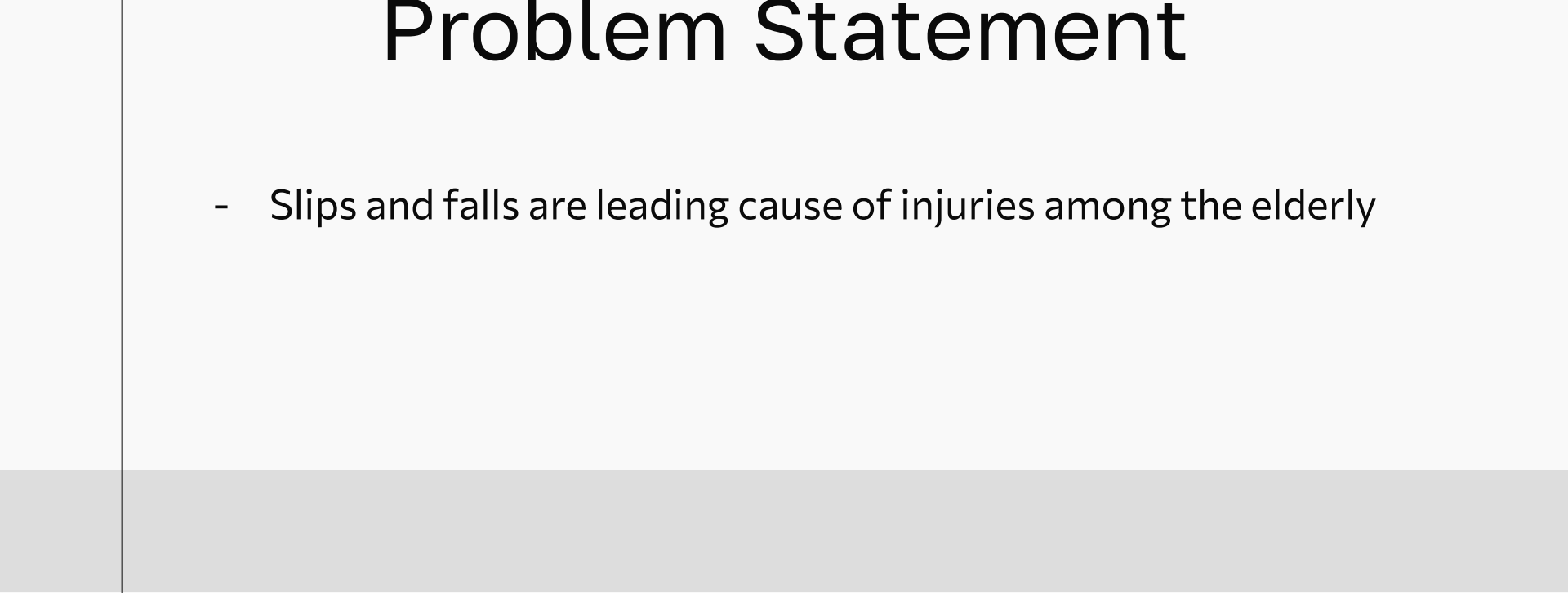
Eric Carty

02

Project Scope

What are we doing here?

Problem Statement

- Slips and falls are leading cause of injuries among the elderly
- 
- A decorative gray horizontal bar spans the bottom of the slide. A thin vertical black line is positioned on the left side, starting from the top and extending to the bottom, intersecting the gray bar.

Our Solution

- Make an hip and knee exoskeleton that prevents falls and slips
 - By tracking the wearer's movements in real time
- And correcting gait perturbations with pneumatic cylinders with use of a gait detection algorithm

03

Deliverables

What have we done?

Deliverables

Deliverable	Description
Rework Design of Thigh Links	— Made new links out of aluminum c-channels — Checked new design with simulations — Reworked thigh brace
IMU System V2	— Improved 400Hz sampling rate over previous versions <200Hz — 0% Data loss during sustained runs — Fully modular Design — UART over RS485 communication protocol
Variable Pneumatic Control	- Improved tuning and testing standards with dedicated test platform — PID Controller program for smooth solenoid actuation

Exoskeleton Old Design

- Thigh linkages were not strong enough to withhold forces
 - Based on tests over winter break



What we changed

- 1) Adapted thigh brace
- 2) Made linkages out of Al c-channels
 - a) With steel support
- 3) Added turnbuckle

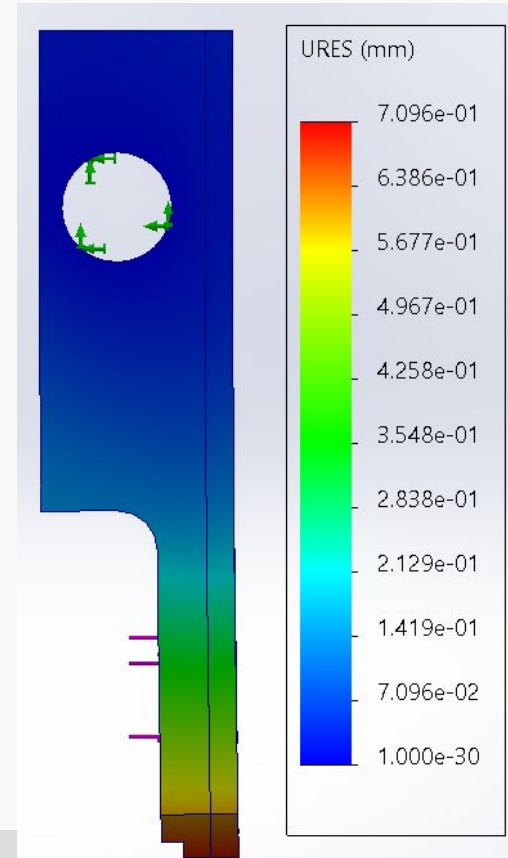
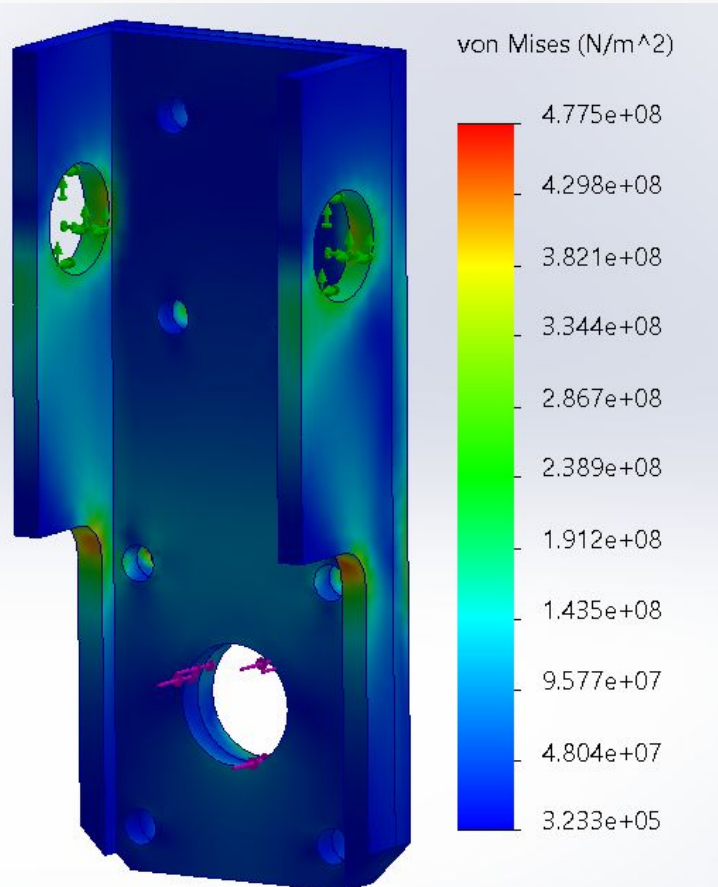
Future Work

Add Flexion capabilities

Add slots for loops on thigh brace



Checking our Thigh Link Design

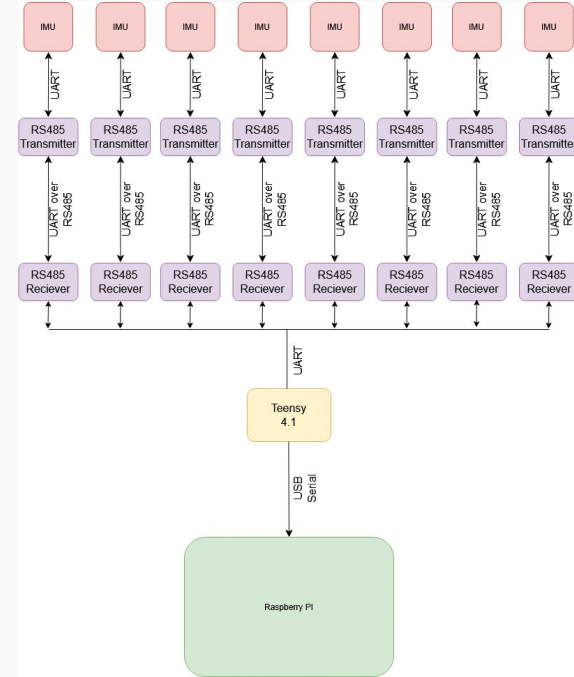


IMU System V2

Key Features:

- UART over RS485 communication protocol prevents potential EMI from high noise electronics such as solenoids and relays
- 400Hz sustained sampling rate with 0% packet loss
- Fully redesigned custom PCB utilizing full duplex transceivers for data reliability over long cable lengths
- Custom made shielded Cat6 cables with twisted pairs and EMI shielding
- Dedicated microcontroller to handle data packets prior to sending them to the Raspberry Pi over serial connection

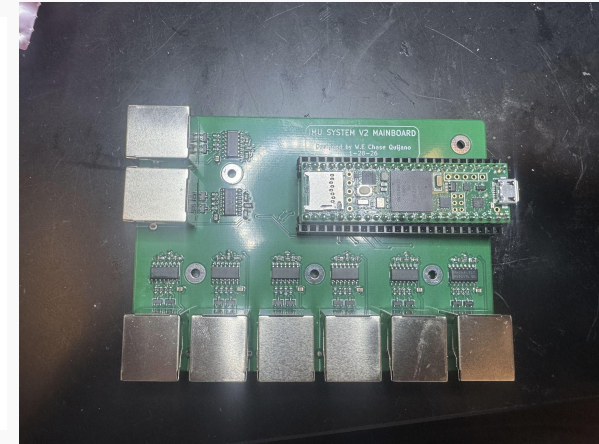
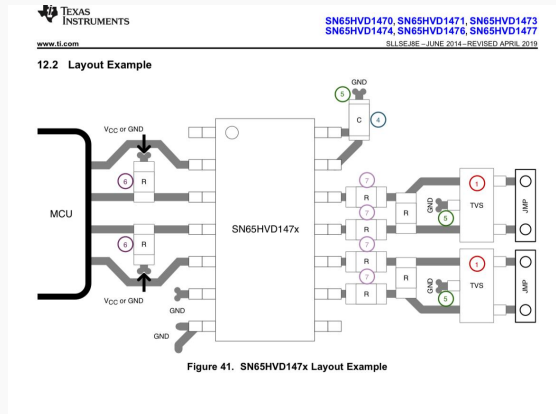
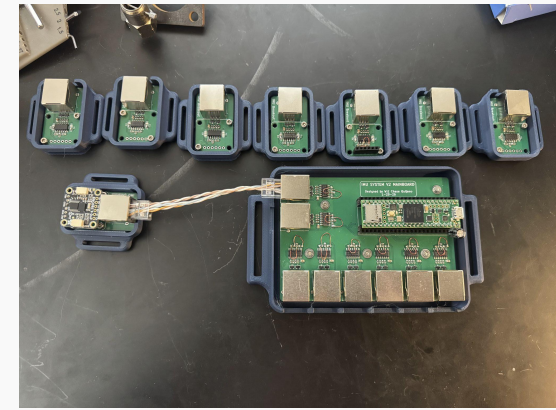
Metric	I2C (Theoretical)	I2C (Real World)	UART with Standard speed
Bus Logic	Sequential	Sequential + Pauses	Parallel
Utilization	61%	183.4%	3.3%
Total Latency	2.34 ms	18.34 ms	0.33 ms
Bandwidth Capacity	427 Hz	54 Hz	19,607 Hz



IMU System V2 - Electrical

Key Features:

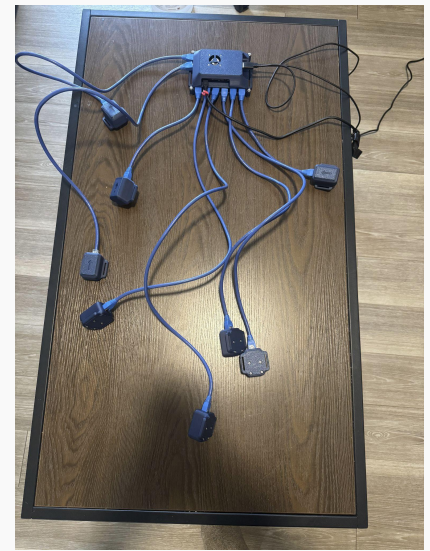
- 7.64ft maximum cable length
- Followed Texas Instruments transceiver design guidelines to ensure signal integrity
- 8 BNO085 IMUs are utilized over UART due to high baseline sampling rate (400Hz)
- Teensy 4.1 is utilized due to high processing speed (700mHz) and 8 dedicated UART channels allowing for simultaneous logging across all 8 IMUs
- RJ45 Jacks used for modularity and commonly available



IMU System V2 - Hardware

Key Features:

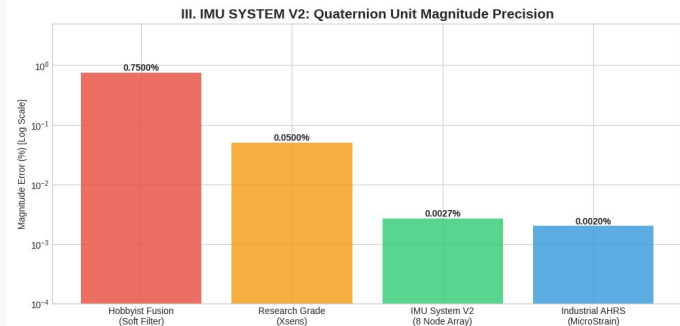
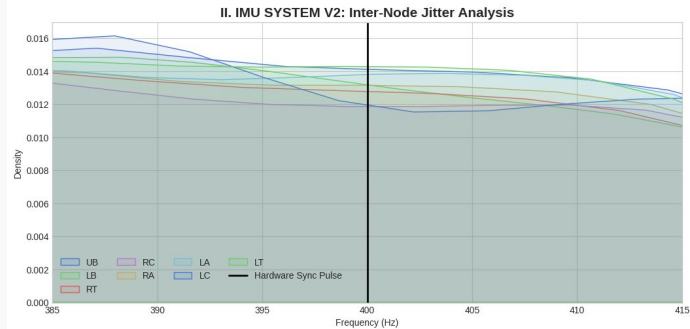
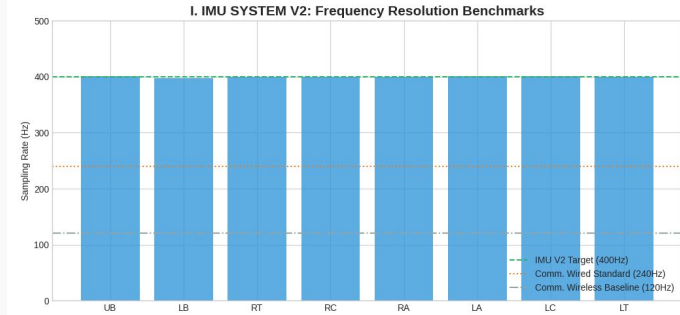
- 3D printed enclosure encompasses Raspberry Pi 4 (Primary computer in charge of gait detection) and IMU System Mainboard
- Low profile and lightweight design for ease of use
- Built in fan that automatically cools system when it passes a temperature threshold
- All ports are accessible on Pi (mini hdmi, power, audio, USB, ethernet) and Teensy (micro-usb) allowing both systems to be programmed or access simultaneously
- IMU Modules have durable enclosure and support custom straps for different body types.



IMU System V2 - DAQ - Unit Test

Performance during 10 minute unit DAQ:

- All nodes consistently stay within 400 Hz range during full duration
- Inter-Node Jitter (The deviation from sample rate between nodes from the target interval (2.5ms). Density is consistently within 400Hz for all 8 IMUs showing consistency in readings)
- Quaternion Unit Magnitude Precision (The deviation of the quaternions magnitude from a unit quaternion during the 10 minute DAQ). Taken from the average across all 8 IMUs Median.
- Reference data derived from: Xsens (Mti - 1), MicroStrain (3DM-GX5-AHRS), Invensense (MPU-6000)



IMU System V2 - DAQ - Worn Test

Key Notes:

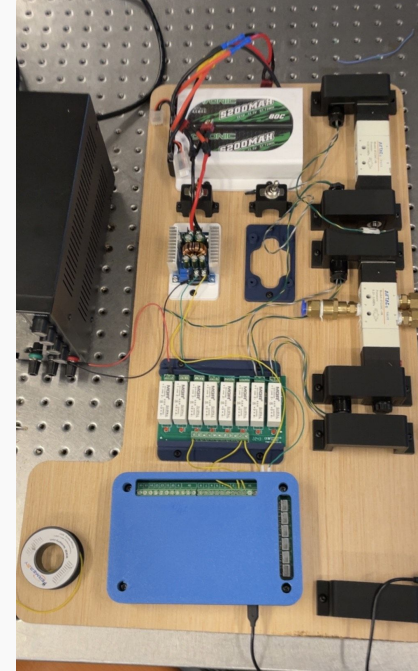
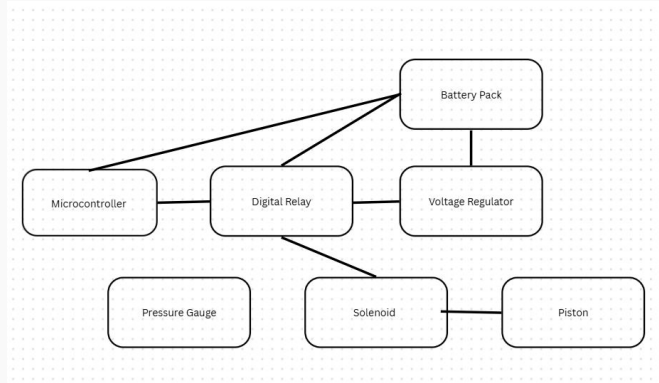
- During a worn test, all 8 IMUs were successfully logged for several minutes
- There was both movement and standstill during the test
- DAQ was done both at 200Hz and 400Hz
- Data is currently under analysis to determine viability



Variable Pneumatic Cylinder Control

By the end of this semester, this module will digitally control the pneumatic piston extension for fall prevention.

- Test Procedure
 - Ideal Parameters
 - Component Verification
- PID Controller
 - Smooth actuation
- Layout and Configuration



Solenoid Tuning Process

In the event of a loss of balance, the device's activation must be comfortable and safe.

- Improvements in Test Procedure
 - Dedicated test bed
 - Improved data collection methods
- Desired Results
 - 1.) How fast will the piston extend at various pressures
 - 2.) How fast will the piston extend at various duty cycles
 - 3.) How does piston extension relate to efficiency losses
- Preliminary Checks
 - Pressure Gauges
 - Lead Wire Connections
 - Voltage Regulation

Thank you!

Any Questions?

04

Appendix

Standards and Regulations

Standard	Description
IEC 60601-1	Safe power distribution, grounding, and insulation
CISPR 11, EN 61000	Ensure no interference with other devices
ISO/IEC 14908	Data Exchange
ISO 5725/ ISO 6789	Maintain sensor precision/calibration
ISO 14791	Risk management and safety evaluations
ISO 13850	Emergency stop/disconnect
Pilz 8.3.3, 8.5.2	Installation of pneumatic linear actuators

*This is only some of the many standards we follow for this project

Muscle Activation During Slip & Slip Recovery



OpenSim

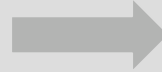
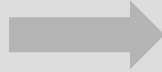
A free software program that allows for the modeling of mechanical and biological structures



Goal

To use data collected from IMU's to create a virtual model of the musculoskeletal system, during slips & recovery.

How It Works



Reading Data

The IMU data is read and converted in OpenSim files

Post Processing

Rotate lab coordinates to match OpenSim coordinate system. Compute force center of pressure

Results

- Perform inverse kinematics
- Perform inverse dynamics
- Use that to calculate which muscles activate and by how much

OpenSim GUI

GUI Components	
Title	Screenshot
Toolbar	
Motion Text Box	
Motion Slider / Video Controls	
View Window	
Navigator Window	
Coordinates Window	
Muscle Editor	

